

FactorizeNet: Progressive Depth Factorization for Efficient Network Architecture Exploration Under Quantization Constraints

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Abstract

Depth factorization and quantization have emerged as two of the principal strategies for designing efficient deep convolutional neural network (CNN) architectures tailored for low-power inference on the edge. However, there is still little detailed understanding of how different depth factorization choices affect the final, trained distributions of each layer in a CNN, particularly in the situation of quantized weights and activations. In this study, we introduce a *progressive depth factorization* strategy for efficient CNN architecture exploration under quantization constraints. By algorithmically increasing the granularity of depth factorization in a progressive manner, the proposed strategy enables a fine-grained, low-level analysis of layer-wise distributions. Thus enabling the gain of in-depth, layer-level insights on efficiency-accuracy tradeoffs under fixed-precision quantization. Such a progressive depth factorization strategy also enables efficient identification of the optimal depth-factorized macroarchitecture design (which we will refer to here as **FactorizeNet**) based on the desired efficiency-accuracy requirements.

1 Introduction

Following the recent explosion in deep learning research, there has been increased attention on complexity reduction strategies for deep convolutional neural networks (CNN) to enable inference on mobile processors. Quantization [1, 2, 3], and depth factorization [4, 5, 6, 7] have quickly emerged as two highly effective strategies for reducing the power and computational budget needed for on-device inference. These two methods work orthogonally. Fixed point quantization enables simple, low bit-width integer operations which are several times faster/less power than floating point (fp32) operations. Depth factorization reduces the number of CNN parameters and multiply-accumulate (MAC) operations. For depth factorization, we split the input channels into f groups and apply f groups of filters independently to their respective channel groups. For a given factorization rate, f , the number of MACs in a convolution layer goes from (1) to (2), thus reducing computation by a factor of f . For simplicity, our equations have excluded the MAC contribution from the pointwise convolution that typically follows the group convolution. Pointwise convolution is often used for the dual purpose of mixing channel information and increasing channel depth.

$$K \times K \times H \times W \times C_{in} \times C_{out} \quad (1)$$

$$K \times K \times H \times W \times \frac{C_{in}}{f} \times \frac{C_{out}}{f} \times f \quad (2)$$

Depthwise separable convolution as described in MobileNets [4] has become a staple in efficient network design. It represents the extreme end of the depth factorization spectrum with one convolution filter per input channel. However, perhaps we do not always need to go to the extreme. A key tradeoff when designing CNNs for limited compute is efficiency vs. accuracy. As we scale down our architectures, we will necessarily lose accuracy. While depthwise separable convolutions are extremely efficient, they suffer from low data parallelism making them less suited to hardware

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